Automated Satirical Football Cartoon Pipeline: Full Technical Specification

Overview: A 100% free, headless cloud pipeline designed to produce 20–35 second vertical (9:16, 1080x1920) daily satirical football cartoon reels for Instagram and YouTube Shorts without manual editing.

1. Core Architecture & Motion Stack

To solve the stiffness of raw 2D cut-outs and avoid open-source video model style-drifting, we use **Programmatic 2D Skeletal Animation (Puppet Rigs)** powered by Python and FFmpeg.

- **Workflow:** Scene JSON Script → Python Parser → Pillow (Asset Compositing & Affine Matrix Transforms) → FFmpeg (Video Render & Audiomuxing).
- **Cloud Execution:** Headless execution via GitHub Actions (CPU runners) or Kaggle Free GPUs (T4, ~30 hrs/wk, 12-hour session limits).

2. Rig-Ready Asset Generation & Layering Standard

Generate modular, transparent-background layers using free image models (e.g., FLUX Schnell) to prevent seams and ensure character consistency:

- **Modular Parts:** `torso.png` (fixed base), `head_neutral.png`, `head_laughing.png`, `head_shocked.png`, `arm_upper_left.png`, `arm_lower_left.png`, `leg_upper.png`, `shoe.png`.
- **Prompt Engineering Rule:** Lock down a master prompt prefix: `"2D vector cartoon caricature, bold black outlines, flat colors, isolated body part on pure white background, no shadows"`. Run output through `rembg` in Python for clean alpha channels.

3. Reusable JSON Scripting & Action Libraries

Actions are driven by predefined animation sequence files stored in a repository library:

- **Action Files:** `walk_cycle.json`, `stamp_papers.json`, and `use_phone.json` containing rotation keyframes and translational offsets for pivots.
- **Renderer Math:** Python calculates affine transformation matrices around designated joint anchor points and pipes frames directly into an in-memory FFmpeg stream.

4. Simplified Lip-Sync & Expression Automation

Skip complex phoneme mapping in favor of a **simplified 3-state mouth sequence** triggered by dialogue timing blocks in the JSON script:

- **State 1:** Closed / Smiling (Idle or listening)
- **State 2:** Open Wide (Shouting punchlines — alternated every 3–4 frames during active speech)
- **State 3:** Smirking / Grinning (Reaction beats)

5. Performance, Cost & Risk Analysis

Parameter	Target / Specification
Resolution & Length	9:16 Vertical, 1080x1920 pixels, 20 to 35 seconds (4–6 scenes).
Render Speed & Cost	1–2 minutes per reel execution; $0.00 total cost using free tier services.
Key Risks & Mitigation	• *Joint Detachment:* Standardize precise anchor point dictionaries. • *Fair Use:* Satirical commentary of public figures is generally protected, avoid commercial defamation.

6. Recommended Stack & 1-Day Test Plan

Recommended Stack: Python (Pillow, numpy, ffmpeg-python) + GitHub Actions + FFmpeg.
1-Day Test Plan: Generate 4 character parts with transparent backgrounds → write a basic Python script to oscillate an arm asset over 60 frames → compile into a 3-second test MP4 via GitHub Actions workflow dispatch.